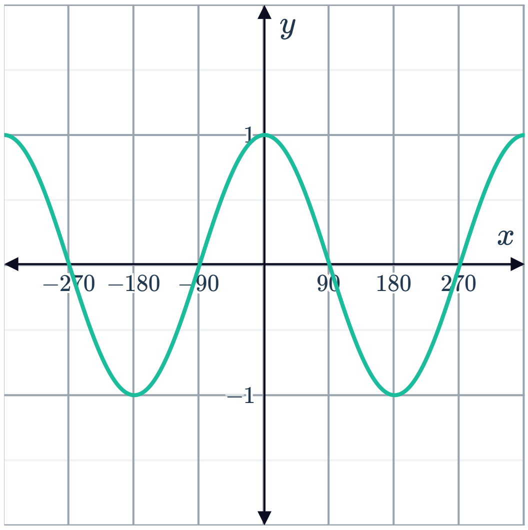
Vertical dilations

1

a

θ	0	60	90	120	180	240	270	300	360
$\cos\theta$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



c 1

d -1

e $[-4, 4]$

2

a $(-\infty, \infty)$

b $[-1, 1]$

3

a $(-\infty, \infty)$

b $[-2, 2]$

4 8

5

a $y = 2\sin x$

b $y = -\sin x$

c $y = 3\cos x$

d $y = -1.5\cos x$

e $y = 3\sin x$

f $y = 2.5\sin x$

6 270

7

a 8

b 0

c -8

8 5

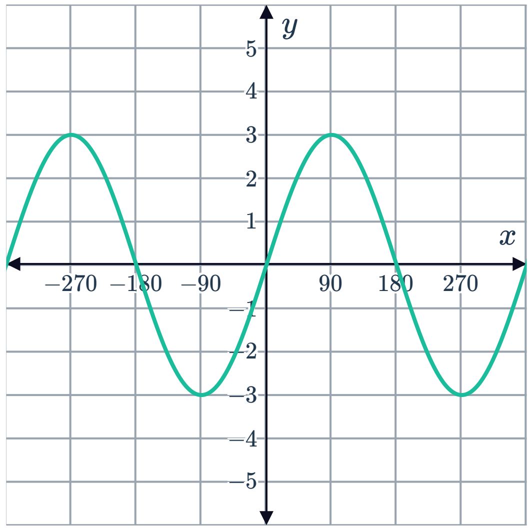
9 $-2\sin x$

10

a

i $A = 3$

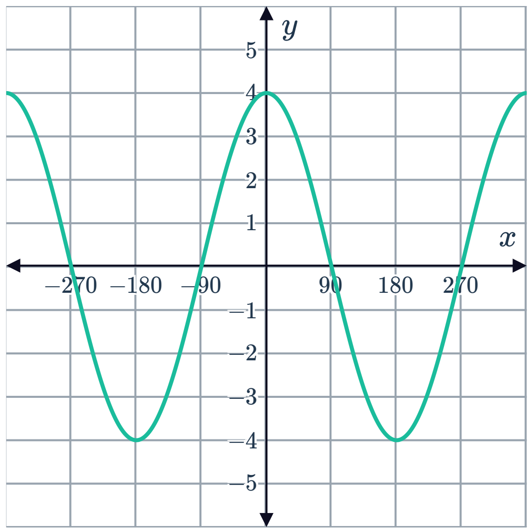
ii



b

i $A = 4$

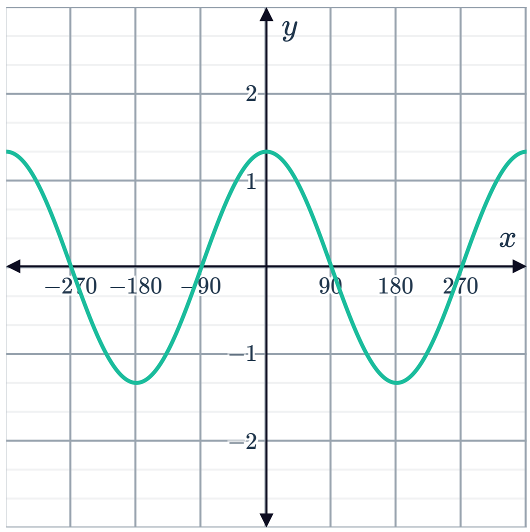
ii



c

i $A = \frac{4}{3}$

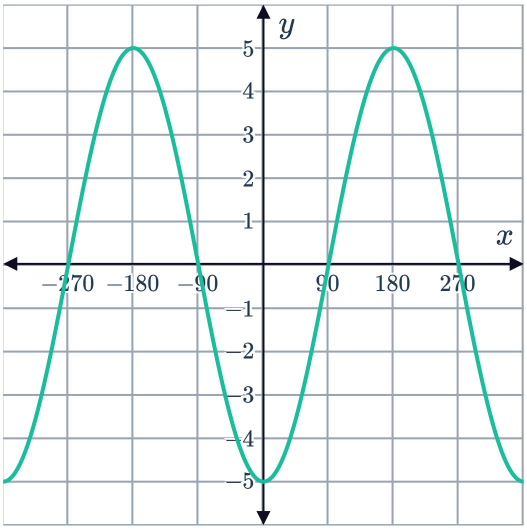
ii



d

i $A = 5$

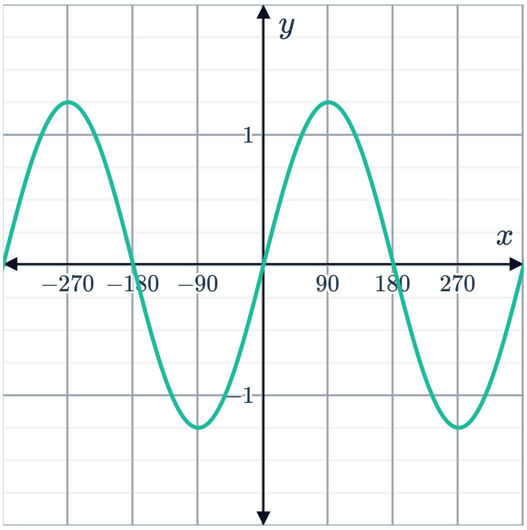
ii



e

i $\frac{5}{4}$

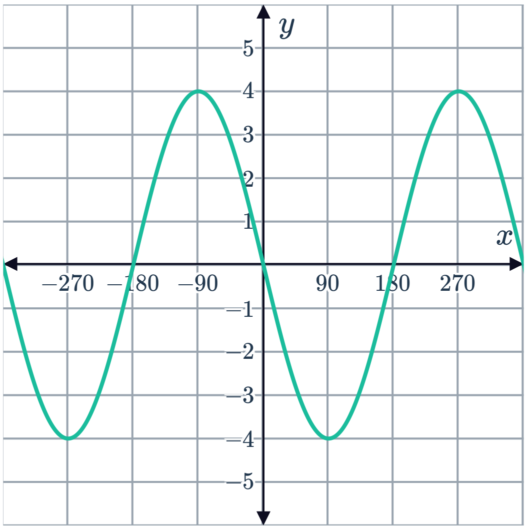
ii



f

i $A = 4$

ii



b -3

c 3

d A vertival dilation by a factor of 3 and reflection about the x-axis.

Vertical translations

12 Translated 2 units downwards.

13 Translated 2 units upwards.

14 Translated 4 units upwards

15

a $y=\sin x+4$

b $y=\sin x-2$

c $y=\sin x-4$

d $y=\sin x+1$

16

a $y=\cos x+3$

b $y=\cos x-4$

c $y=\cos x-3$

d $y=\cos x+2$

17 Reflection about the x-axis, then translation 3 units up.

18 $y=2$

19

a Translated 4 units upwards.

b 360°

c 5

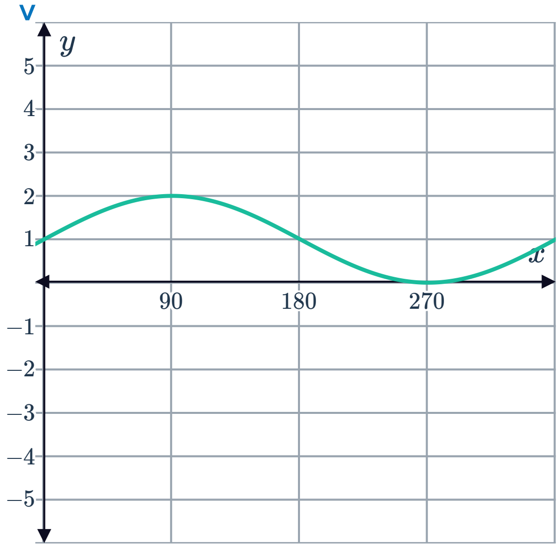
d 3

20

a

i 360 **ii** 1 **iii** 2 **iv** 0

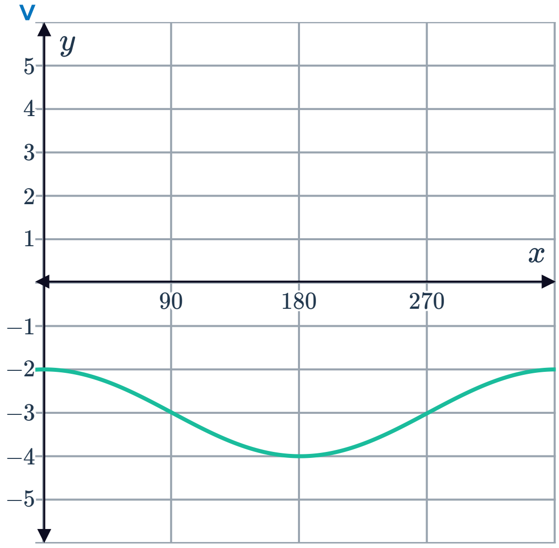
v



b

i 360 **ii** 1 **iii** -2 **iv** -4

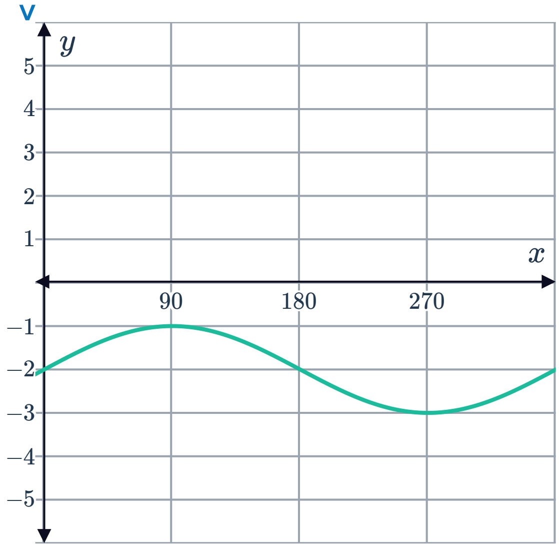
v



c

i 360 **ii** 1 **iii** -1 **iv** -3

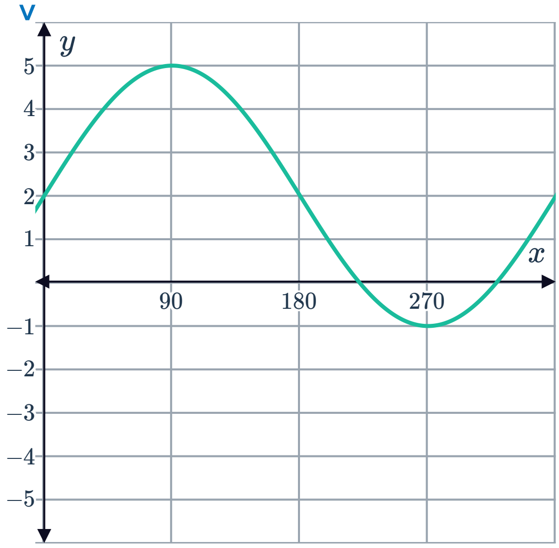
v



d

i 360 **ii** 3 **iii** 5 **iv** -1

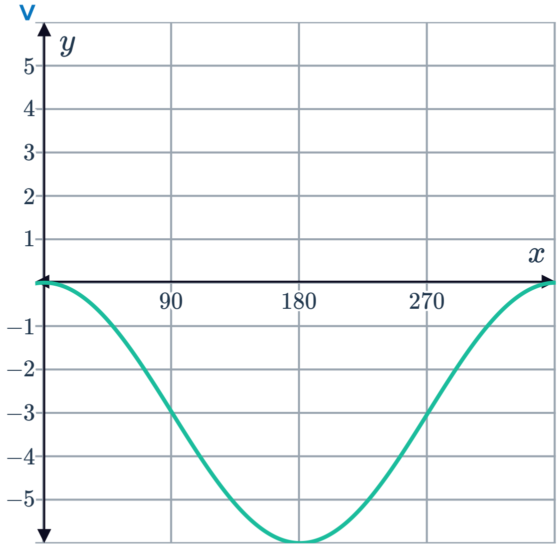
v



e

i 360 **ii** 3 **iii** 0 **iv** -6

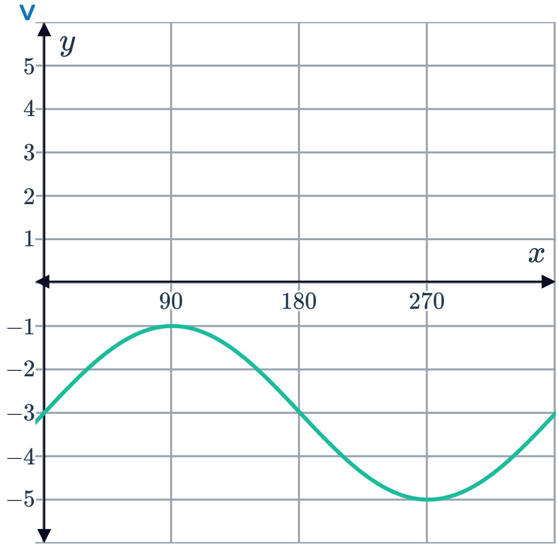
v



f

i 360 **ii** 2 **iii** -1 **iv** -5

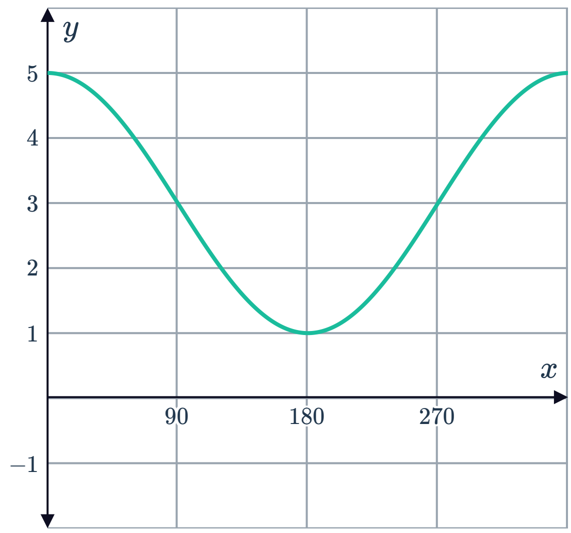
v



21

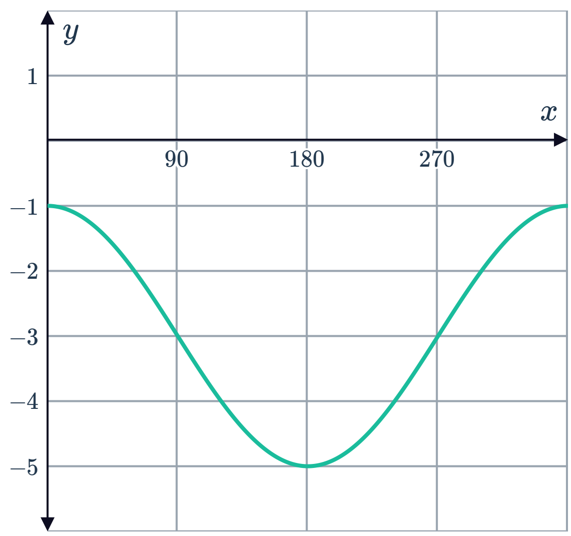
a

- **i** 3
- **ii** 2
- **iii** 5
- **iv** 1
- **v** A vertical dilation by a factor of 2, then translated 3 units upwards.
- **vi**



b

- **i** -3
- **ii** -1
- **iii** -5
- **iv** 2
- **v** A vertical dilation by a factor of 2, then translated 3 units downwards.
- **vi**



c

i 2

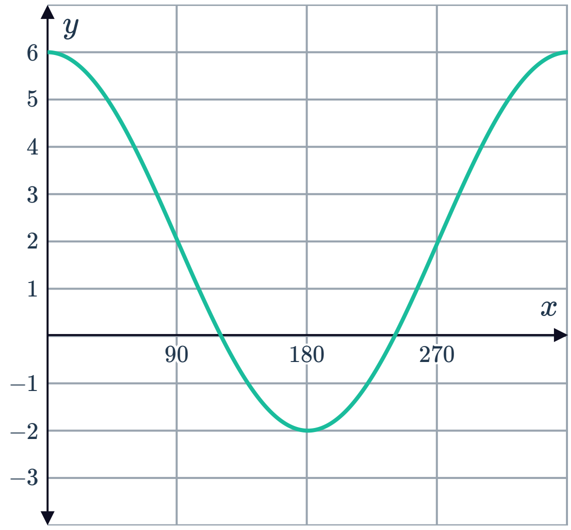
ii 6

iii -2

iv 4

v A vertical dilation by a factor of 4, a reflection about the x-axis and then translated 2 units upwards.

vi



d

i 2

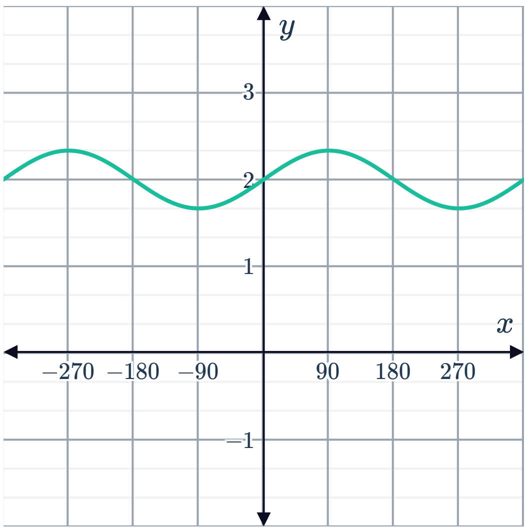
ii $\frac{7}{3}$

iii $\frac{5}{3}$

iv $\frac{1}{3}$

v A vertical dilation by a factor of $\frac{1}{3}$, then translated 2 units upwards.

vi



22 **a** $y = \cos x + 9$ **b** 10

23

a **i** 5 **ii** -1

b **i** 7 **ii** 1

c **i** 2 **ii** 1

d **i** 2 **ii** $\frac{1}{2}$

24 $2 \sin x + 2$

25 $8 \cos x - 2$